Momentum and Newton’s laws of motion

AS Level | 9702 AS Ch.6 | Chapter 6: Momentum | Lesson 6.6 | Sections 6.6

Goal: Use force = rate of change of momentum, including for collisions and streams of fluid.

01 The big picture

Newton’s second law in its general form: the resultant force on an object equals its rate of change of momentum. For a constant mass, $\Delta p/\Delta t = m\,\Delta v/\Delta t = ma$.

Rearranged, $F\Delta t = \Delta p$: the same change in momentum over a longer time needs a smaller force. That is why crumple zones, airbags and bending your knees on landing protect you.

Force (Newton’s second law) [paraphrase]
The resultant force on an object equals its rate of change of momentum.

02 Key equations

Newton’s second law

$$F = \frac{\Delta p}{\Delta t}$$

Constant mass

$$F = ma$$

Rearranged

$$F\,\Delta t = \Delta p$$

Symbols and units

F	resultant force	N	Δp	change in momentum	$\mathrm{kg\,m\,s^{-1}}$	Δt	time taken	s

03 Sample problems with solutions

Sample 1 · Hitting a ball

A 0.15 kg ball arrives at a bat at 20 m s⁻¹ and leaves in the opposite direction at 30 m s⁻¹. Contact lasts 0.010 s. Calculate the average force on the ball.

Step 1. Return direction positive, so $u = -20$ m s⁻¹:

$$\Delta p = 0.15\,(30 - (-20)) = 7.5\ \mathrm{kg\,m\,s^{-1}}$$

Step 2. Rate of change of momentum:

$$F = \frac{7.5}{0.010} = 750\ \mathrm{N}$$

Answer: 750 N, in the direction the ball leaves

Unit check: $\frac{\mathrm{kg\,m\,s^{-1}}}{\mathrm{s}} = \mathrm{kg\,m\,s^{-2}} = \mathrm{N}$ ✓

04 Practice problems

P1 A 1000 kg car goes from rest to 20 m s⁻¹ in 8.0 s. Find the average resultant force.
Answer: 2500 N

P2 A hose sends 2.0 kg of water per second at 15 m s⁻¹ onto a wall, where it stops. Find the force on the wall.
Answer: 30 N

P3 A momentum change of 12 kg m s⁻¹ takes 0.040 s. Find the average force.
Answer: 300 N

P4 Why do you bend your knees when you land from a jump?
Answer: Longer stopping time, so a smaller force for the same Δp

05 Other useful information

Common mistakes

✗ **Wrong:** Use $F = ma$ for a stream of fluid.
✓ **Right:** Use $F = \Delta p/\Delta t$: mass per second × change in velocity.

Exam tips

- Check units: kg m s⁻¹ per second is kg m s⁻², which is N.

What the syllabus asks

- 9702 topic 3: define and use force as rate of change of momentum. Coursebook 6.6.

Explore: Practice questions for this lesson